notebook1

The purpose of this notebook is to document the development of the analysis. The final analysis will be run through a series of scripts, but I will develop those scripts interactively here and document my decision making process as I go.

```
# since this notebook is not in the main directory, use the here package
# to get proper file paths
library(here)
```

```
here() starts at /mnt/c/Users/jjk06/OneDrive/Desktop/2026_r_projs/hybrid_fitness_predict
```

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Creating a file of seed sizes

Seeds were planted 20 at a time into grids of 4 rows by 5 columns. Before planting the seeds, they were placed onto a grid as they were to be planted and a photo taken with a scale in the photo. The areas of the seeds were then measured from the photograph. Four researchers were working on the project, to get used to using imageJ and make sure there was no bias, the seeds were measured by all four researchers.

First, I'm reading in the four files and just calculate the mean seed area for each seed.

```
initials <- c("jk", "js", "lb", "rp")

# there were some notes etc. beyond the 4th column,
# so only take the first 4 columns
tmp <- read.csv(here(paste0("data/raw/seed_areas_",
                            initials[1], ".csv")))[,1:4]

tmp$researcher <- initials[1]

d <- tmp

for(i in 2:length(initials)){
  tmp <- read.csv(here(paste0("data/raw/seed_areas_",
                              initials[i], ".csv")))[,1:4]

  tmp$researcher <- initials[i]

  d <- rbind(d, tmp)
}


# calculate the mean of the four measurements
seed_size <- d %>%
  group_by(tray, block, ind) %>%
  summarise(seed_area = round(mean(area),3))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, and ind.
i Output is grouped by tray and block.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Next, I am going to attach the cross id and parentage to each individual.

```
ids <- read.csv(here("data/raw/planting_ids.csv"))

# tray format is different in the two files, so need to change that to merge
ids <- ids %>%
  mutate(tray = gsub("LC", "lc_", tray)) %>%
  select(tray, block, cross, mother, father)

seed_size <- seed_size %>%
  left_join(ids) %>%
  mutate(cross = tolower(cross))
```

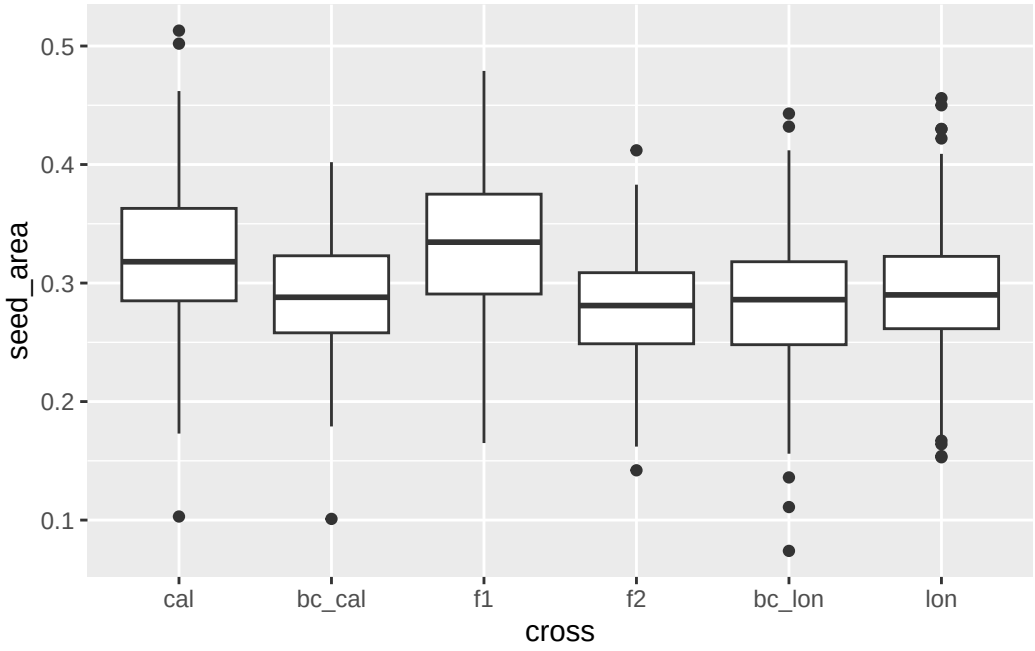
```
Joining with `by = join_by(tray, block)`
```

The above code takes the generates an object with seed sizes and cross information. It will be used in the script `scripts/wrangling.R` to generate the intermediate data file `reulsts/data/seed_size.csv`

Exploratory data analysis with seed size

First let's look at seed size vs cross type.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = cross, y = seed_area)) +
  geom_boxplot()
```



It looks like f1 and cal are the biggest with everything else being fairly similar. Let's just do a quick anova to see what that looks like.

```
summary(mod <- aov(seed_area ~ cross, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
cross          5  0.594 0.11876   41.11 <2e-16 ***
Residuals   1370  3.957 0.00289
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

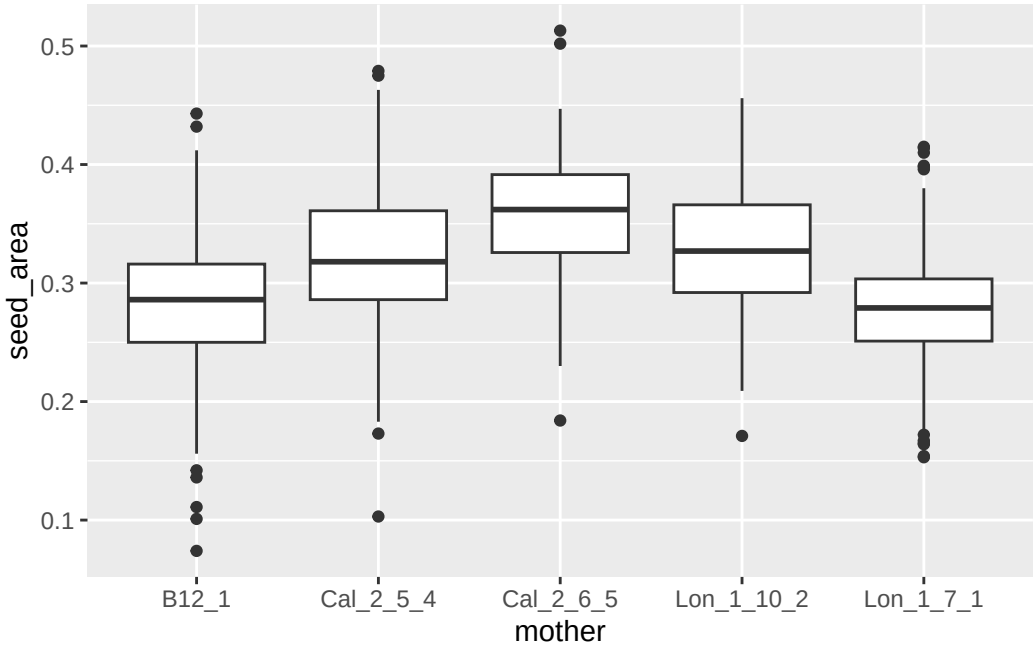
Fit: aov(formula = seed_area ~ cross, data = seed_size)

$cross
                       diff           lwr          upr     p adj
bc_lon-bc_cal -0.004700301 -0.0190972135  0.009696612 0.9383445
```

```
cal-bc_cal     0.036615225  0.0223096875  0.050920762 0.0000000
f1-bc_cal      0.044222433  0.0298717261  0.058573140 0.0000000
f2-bc_cal     -0.009266329 -0.0237426551  0.005209996 0.4487747
lon-bc_cal     0.004501934 -0.0098640582  0.018867925 0.9479315
cal-bc_lon     0.041315525  0.0270739589  0.055557092 0.0000000
f1-bc_lon      0.048922734  0.0346357954  0.063209672 0.0000000
f2-bc_lon     -0.004566029 -0.0189791416  0.009847085 0.9455309
lon-bc_lon     0.009202234 -0.0051000571  0.023504526 0.4427199
f1-cal         0.007607208 -0.0065876466  0.021802063 0.6452916
f2-cal        -0.045881554 -0.0602033949 -0.031559713 0.0000000
lon-cal       -0.032113291 -0.0463235986 -0.017902983 0.0000000
f2-f1         -0.053488762 -0.0678557217 -0.039121803 0.0000000
lon-f1        -0.039720499 -0.0539762784 -0.025464720 0.0000000
lon-f2         0.013768263 -0.0006139641  0.028150490 0.0697293
```

So it seems like there might be some real differences here. I'm not so sure that it is because of the hybrid embryo though. Seed size is at least partially, if not mostly a maternal trait, so lets look at seed size vs seed parent.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area)) +
  geom_boxplot()
```



Looks like there are some differences, let's do a quick anova and see.

```
summary(mod <- aov(seed_area ~ mother, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
mother         4  0.814 0.20339   74.61 <2e-16 ***
Residuals   1371  3.737 0.00273
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

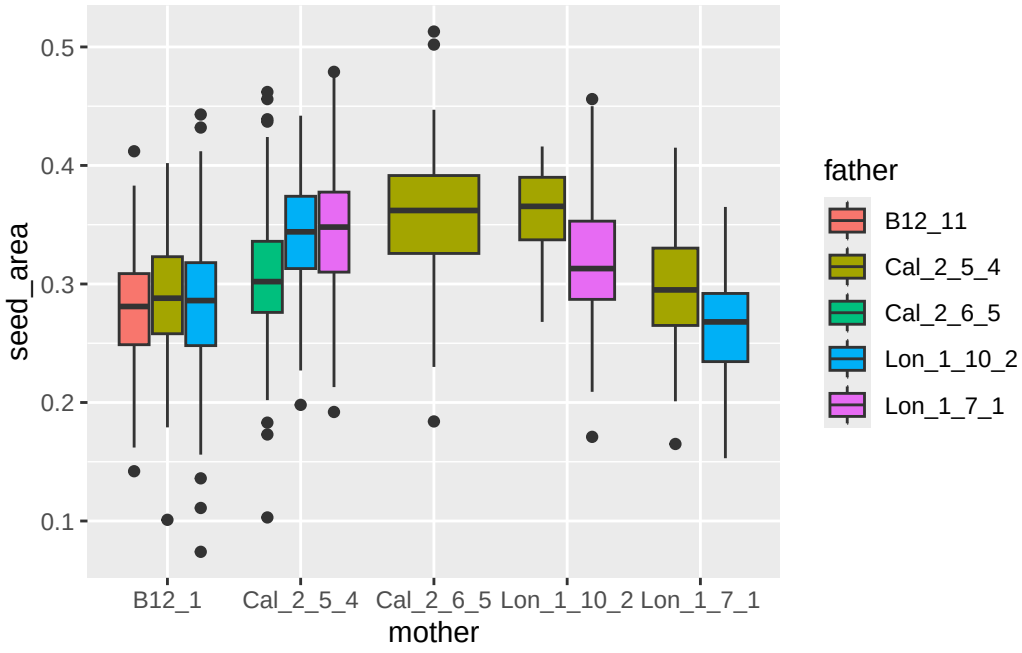
Fit: aov(formula = seed_area ~ mother, data = seed_size)

$mother
                            diff          lwr          upr     p adj
Cal_2_5_4-B12_1       0.039812279  0.029589976  0.050034582 0.0000000
Cal_2_6_5-B12_1       0.073819605  0.056768466  0.090870745 0.0000000
```

```
Lon_1_10_2-B12_1       0.046663473  0.034032604  0.059294341 0.0000000
Lon_1_7_1-B12_1       -0.004230057 -0.015936444  0.007476330 0.8612564
Cal_2_6_5-Cal_2_5_4    0.034007326  0.015697695  0.052316957 0.0000044
Lon_1_10_2-Cal_2_5_4   0.006851193 -0.007433066  0.021135453 0.6849996
Lon_1_7_1-Cal_2_5_4   -0.044042337 -0.057516042 -0.030568632 0.0000000
Lon_1_10_2-Cal_2_6_5  -0.027156133 -0.046911789 -0.007400476 0.0016930
Lon_1_7_1-Cal_2_6_5   -0.078049663 -0.097227426 -0.058871899 0.0000000
Lon_1_7_1-Lon_1_10_2  -0.050893530 -0.066274820 -0.035512240 0.0000000
```

It looks like there might be some real differences between the mothers. Most of the parentals were crossed with an opposite parent as well, so let's add in paternal information to see how the embryo genotype affects seed size.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area, fill = father)) +
  geom_boxplot()
```



It seems that making crosses between parents increases seed size within the same mother, i.e., f1 embryos produce bigger seeds. The F1 mother produces the same size seeds whether crossed with another f1, lon, or, cal.

One last thing to do with seed size before moving on is that there is some missing data here. What happened is we tried to pre-apportion 20 seeds for each block ahead of time, but when we poured out the seeds or were arranging them to be photographed, we would sometimes loose a couple as they would jump around from static electricity or a misplaced sigh and then couldn't find them again. When that happened, we would leave the last places blank. So I just want to go through and figure out how many were missing from each block. Just placing it here so If I need it for quick reference later I can come find it.

```
seed_size %>%
  drop_na(seed_area) %>%
  group_by(tray, block) %>%
  summarise(n = n())
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray and block.
i Output is grouped by tray.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 72 x 3
# Groups:   tray [6]
   tray  block     n
   <chr> <int> <int>
 1 lc_1      1    15
 2 lc_1      2    20
 3 lc_1      3    17
 4 lc_1      4    20
 5 lc_1      5    20
 6 lc_1      6    20
 7 lc_1      7    20
 8 lc_1      8    20
 9 lc_1      9    20
10 lc_1     10    20
# i 62 more rows
```

Wrangling the germination timing data

I think the next set of data that needs to be wrangled is germination timing. The file `data/raw/germination_timing.csv` has observations of germination. The column names here are tray, block, ind, which correspond with the seed areas above. The rest of the columns are days that I checked for germination with the format 24263_16 (R adds an X that I'll have to remove). Where 24263 means the Julian date 263 for year 2024 and the 16 indicates the time the observation was taken. I think the time is probably too fine grained for this analysis, so I am going to remove it as well. For efficiency's sake, I entered a 1 if the plant had germinated and didn't enter anything if it hadn't, so no germination at a time point will be an NA

```
germ <- read.csv(here("data/raw/germination_timing.csv"))
head(germ)
```

```
  tray block ind X24263_16 X24264_17 X24265_14.5 X24266_14.5 X24267_13.5
1 lc_1     1   1        NA        NA          NA          NA          NA
2 lc_1     1   2        NA        NA          NA          NA          NA
3 lc_1     1   3        NA        NA          NA          NA          NA
4 lc_1     1   4        NA        NA          NA          NA          NA
5 lc_1     1   5        NA        NA          NA          NA          NA
6 lc_1     1   6        NA        NA          NA          NA          NA
  X24269_10 X24270_15.5 X24271_13.5 X24274_16.25 X24277_13.0
1         1           1           1            1           1
2        NA           1           1            1           1
3         1           1           1            1           1
4         1           1           1            1           1
5        NA          NA          NA           NA           1
6        NA          NA          NA           NA          NA
```

First thing is to make it a little more tidy.

```
(germ <- germ %>%
  pivot_longer(4:ncol(germ), names_to = "date", values_to = "germ") %>%
  # get rid of X in front of date, separate and get rid of the time
  # and make date numeric (also get rid of 24 since they are all
  # 2024)
  mutate(date = gsub("X24", "", date)) %>%
  separate(date, into = c("date", NA), sep = "_") %>%
  mutate(date = as.numeric(date)) %>%
  # now drop all the NAs when a germ wasn't observed and keep only the
  # earliest observed date, i.e., when it germinated
```

```
  drop_na(germ) %>%
  group_by(tray, block, ind) %>%
  filter(date == min(date)))
```

```
# A tibble: 904 x 5
# Groups:   tray, block, ind [904]
   tray  block   ind  date  germ
   <chr> <int> <int> <dbl> <int>
 1 lc_1      1     1   269     1
 2 lc_1      1     2   270     1
 3 lc_1      1     3   269     1
 4 lc_1      1     4   269     1
 5 lc_1      1     5   277     1
 6 lc_1      1     7   274     1
 7 lc_1      1     8   270     1
 8 lc_1      1     9   269     1
 9 lc_1      1    10   269     1
10 lc_1      1    11   274     1
# i 894 more rows
```

There is one more source of germination timing in the form of the seedling areas. We measured the seedling areas through the time period that germination was being taken, so if there is a measurement for the seedling it must have germinated. I'll blend these together to make sure I didn't miss anything.

```
areas <- read.csv(here("data/raw/seedling_areas.csv"))
head(areas)
```

```
  tray block ind area       time observer notes
1    1     1   1    1      24267_14.5       js
2    1     1   2    2      24267_14.5       js
3    1     1   3    3      24267_14.5       js
4    1     1   4    4      24267_14.5       js
5    1     1   5    5      24267_14.5       js
6    1     1   6    6      24267_14.5       js
```

The time column here is when the seedling area was measured. Again, I'm going to drop the hours and get rid of the 24. It looks like area was read in as a character, so I'm going to force it into numeric then drop the NAs

```r
(areas <- areas %>%
  # remove the 24 and hours
  separate(time, into = c("day", NA), sep = "_") %>%
  mutate(day = gsub("24","", day),
         day = as.numeric(day)) %>%
  # make area numeric and drop NAs
  mutate(area = as.numeric(area)) %>%
  drop_na(area) %>%
  # now get the earliest day for each individual
  group_by(tray, block, ind) %>%
  filter(day == min(day)) %>%
  # since multiple researchers measured each day/seedling combo, there
  # are multiple instances of each day, so filter down to just one
  ungroup() %>%
  filter(!duplicated(paste0(tray, block, ind))))
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `area = as.numeric(area)`.
Caused by warning:
! NAs introduced by coercion
```

```
# A tibble: 902 x 7
    tray block   ind  area   day observer notes
   <int> <int> <int> <dbl> <dbl> <chr>    <chr>
 1     1     1    14 1.27    267 js       ""
 2     1     2     3 2.31    267 js       ""
 3     1     2    12 0.558   267 js       ""
 4     1     3     4 3.16    267 js       ""
 5     1     3     5 1.49    267 js       ""
 6     1     3     8 3.31    267 js       ""
 7     1     3    14 1.09    267 js       ""
 8     1     4     3 4.26    267 js       ""
 9     1     4     4 2.58    267 js       ""
10     1     4     6 2.54    267 js       ""
# i 892 more rows
```

Now I'm going to join the two data sets and select the earliest observation.

```r
(germ_day <- areas %>%
  mutate(tray = paste0("lc_", tray)) %>%
  left_join(germ, by = join_by(tray, block, ind)) %>%
```

```
  pivot_longer(c(day, date), values_to = "germ_day") %>%
  drop_na(germ_day) %>%
  group_by(tray, block, ind) %>%
  filter(germ_day == min(germ_day)) %>%
  filter(!duplicated(tray, block, ind)) %>%
  select(tray, block, ind, germ_day))
```

```
# A tibble: 902 x 4
# Groups:   tray, block, ind [902]
   tray  block   ind germ_day
   <chr> <int> <int>    <dbl>
 1 lc_1      1    14      265
 2 lc_1      2     3      265
 3 lc_1      2    12      266
 4 lc_1      3     4      263
 5 lc_1      3     5      266
 6 lc_1      3     8      264
 7 lc_1      3    14      265
 8 lc_1      4     3      263
 9 lc_1      4     4      265
10 lc_1      4     6      265
# i 892 more rows
```

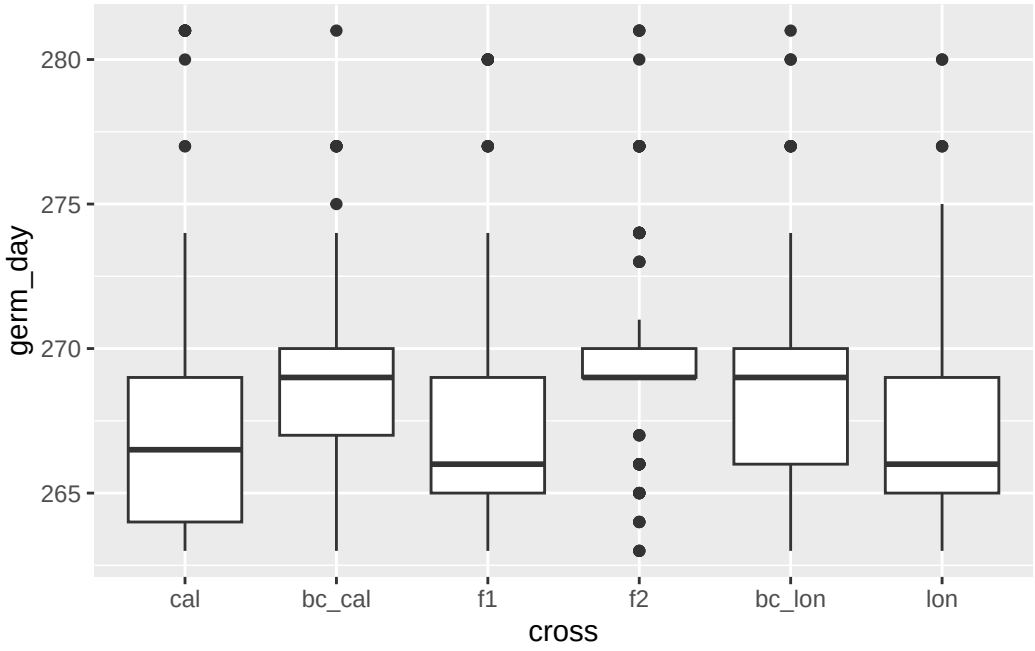
Exploratory analysis of germination timing

Ok, now that we have some germ times, let's join it with the planting ids to get cross and parentage info and do some exploratory analysis.

First, let's look at germination time vs cross.

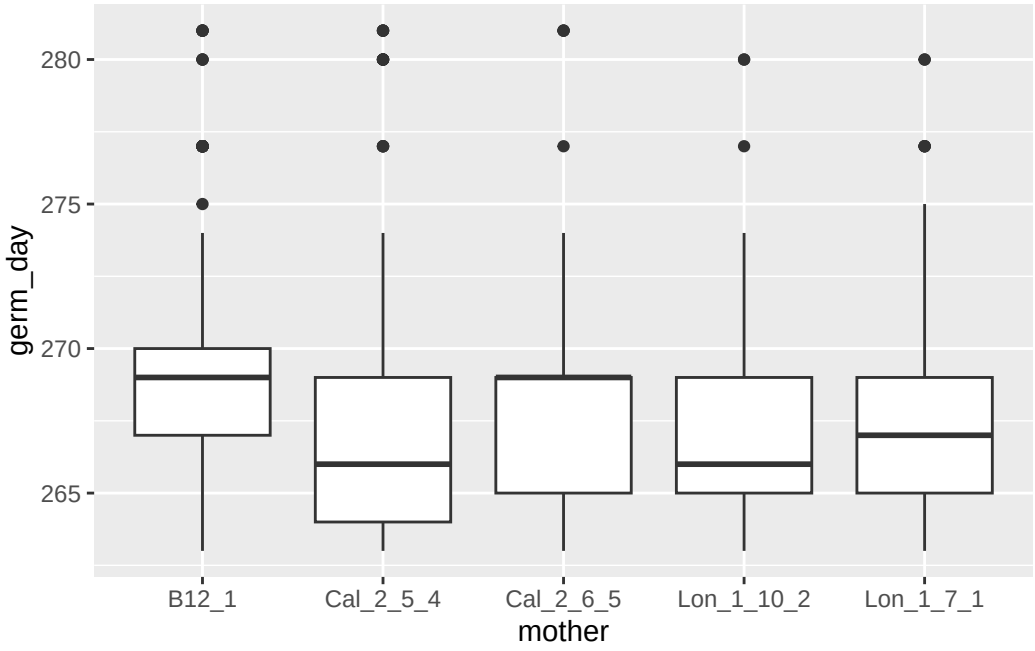
```
germ_day <- germ_day %>%
  left_join(ids, by = join_by(tray, block)) %>%
  mutate(cross = tolower(cross),
         cross = factor(cross,
                        levels = c("cal", "bc_cal", "f1",
                                   "f2", "bc_lon", "lon")))

germ_day %>%
  ggplot(aes(x = cross, y = germ_day)) +
  geom_boxplot()
```



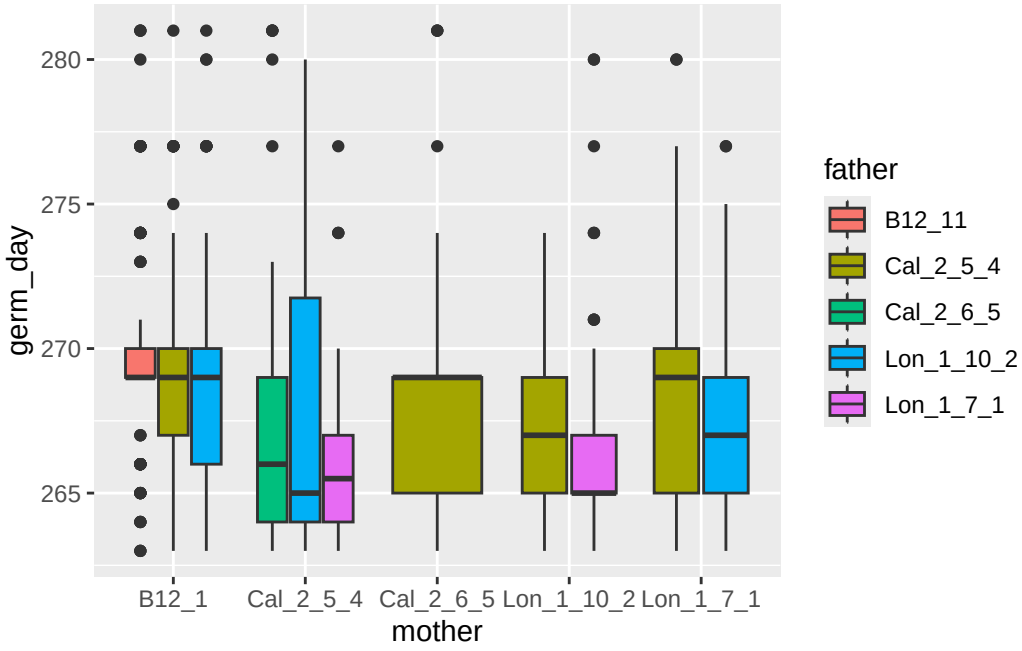
It looks like the back crosses and f2 take longer to germinate than the f1s and parentals. Let's look at it by seed parent.

```
germ_day %>%
  ggplot(aes(x = mother, y = germ_day)) +
  geom_boxplot()
```



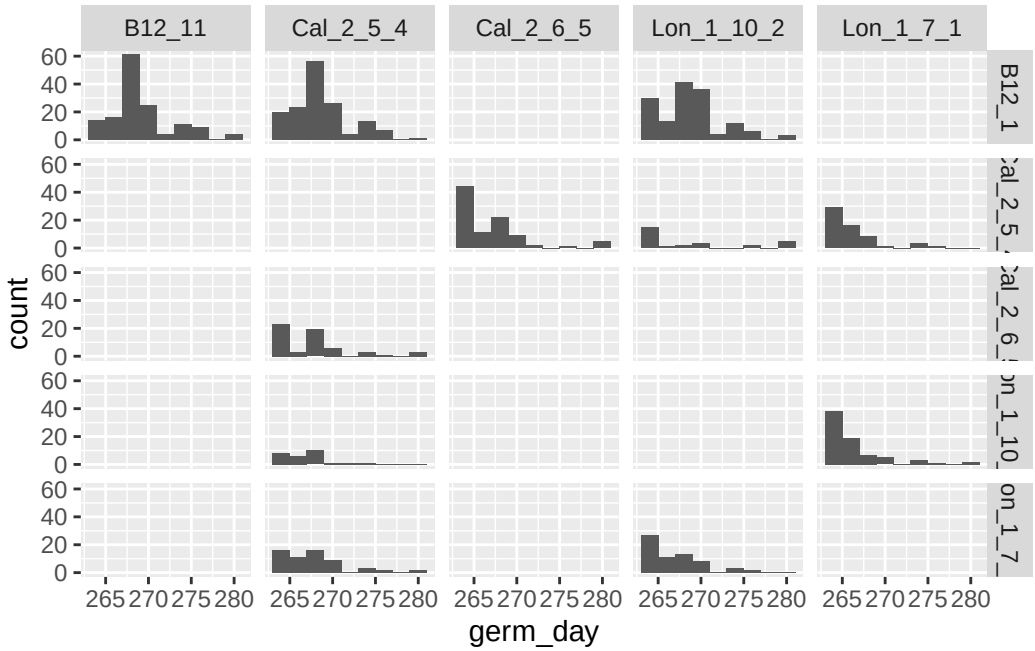
Not sure what to think abut this, but let's look at it breaking out the individual crosses (i.e. include the father information)

```
germ_day %>%
  ggplot(aes(x = mother, y = germ_day, fill = father)) +
  geom_boxplot()
```



The f2 pattern looks really weird so let's look at the distribution of germ days.

```
germ_day %>%
  ggplot(aes(x = germ_day)) +
  geom_histogram(binwidth = 2) +
  facet_grid(mother ~ father)
```



Is there a relationship between seed size and germination timing?

```
seed_size <- seed_size %>% ungroup()
germ_day <- germ_day %>% ungroup()

d <- seed_size %>%
  left_join(germ_day)
```

```
Joining with `by = join_by(tray, block, ind, cross, mother, father)`
```

```
# since the experiment started on Julian day 260, subtract that from germ
# day to get the actual germ_day.

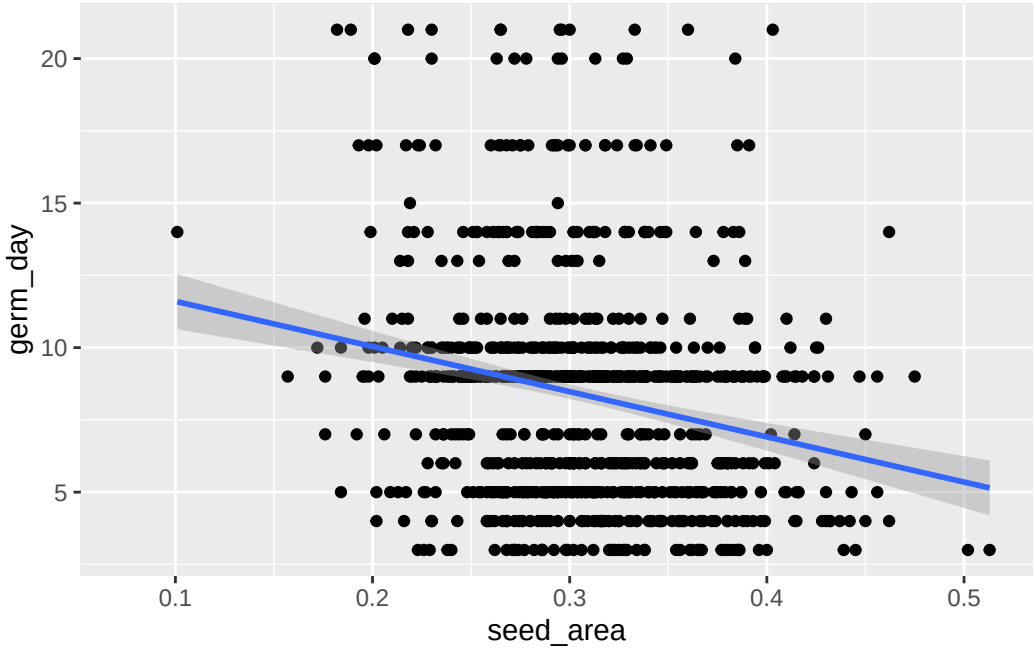
d <- d %>%
  mutate(germ_day = germ_day - 260)

d %>%
  ggplot(aes(x = seed_area, y = germ_day)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 540 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 540 rows containing missing values or values outside the scale range
(`geom_point()`).
```



There definitely seems to be a negative relationship between seed size and germination timing.

I'm going to add some of the above code to the `wrangling.R` script so that the output includes germination timing as well.

Probability of Germination

Next I want to look at the probability of germination and check out the relationship between size and cross type.

```
(d <- d %>%
  # first drop the NA in seed_area because they were places where
  # there are no seeds.
  drop_na(seed_area) %>%
  # convert germ day into whether or not a plant germinated. NAs are
  # non-germinants
```

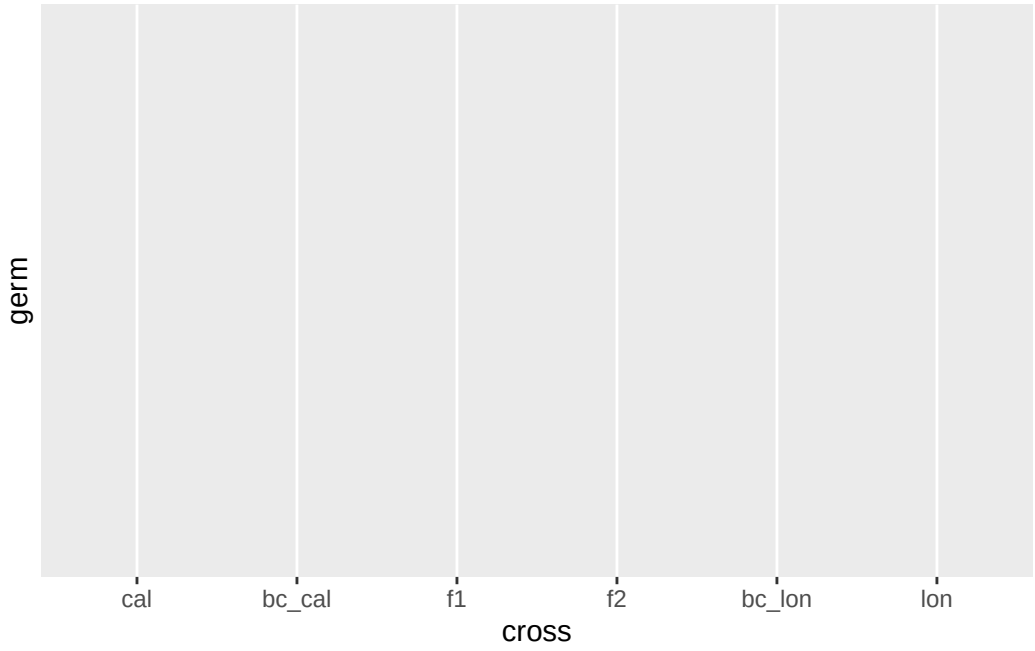
```
  mutate(germ = ifelse(is.na(germ_day), 0, 1)) %>%
  mutate(cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2",
                                          "bc_lon", "lon"))))
```

```
# A tibble: 1,376 x 9
   tray  block   ind seed_area cross mother father germ_day  germ
   <chr> <int> <int>     <dbl> <fct> <chr>  <chr>     <dbl> <dbl>
 1 lc_1      1     1     0.322 f2    B12_1  B12_11        9     1
 2 lc_1      1     2     0.287 f2    B12_1  B12_11       10     1
 3 lc_1      1     3     0.286 f2    B12_1  B12_11        9     1
 4 lc_1      1     4     0.274 f2    B12_1  B12_11        9     1
 5 lc_1      1     5     0.318 f2    B12_1  B12_11       17     1
 6 lc_1      1     6     0.239 f2    B12_1  B12_11       NA     0
 7 lc_1      1     7     0.253 f2    B12_1  B12_11       14     1
 8 lc_1      1     8     0.228 f2    B12_1  B12_11       10     1
 9 lc_1      1     9     0.33  f2    B12_1  B12_11        9     1
10 lc_1      1    10     0.313 f2    B12_1  B12_11        9     1
# i 1,366 more rows
```

Let's look at germination by cross first.

```
d %>%
  ggplot(aes(x = cross, y = germ)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```

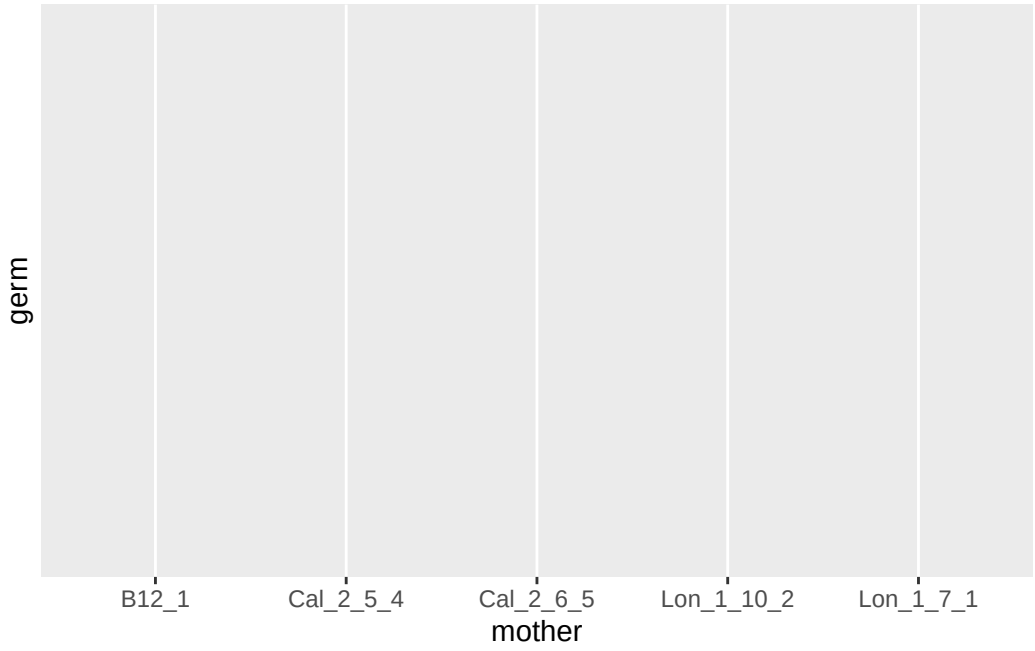


It looks like the f1s have a higher germination percentage than the rest, but not so different between everything else maybe.

Let's look at by mother now.

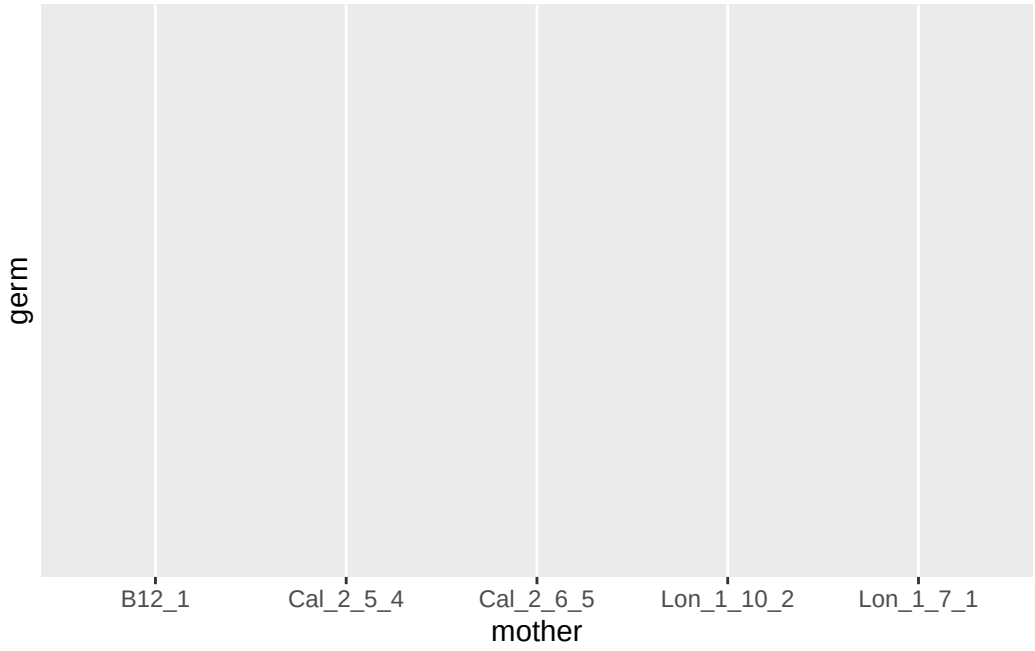
```
d %>%
  ggplot(aes(x = mother, y = germ)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar")
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



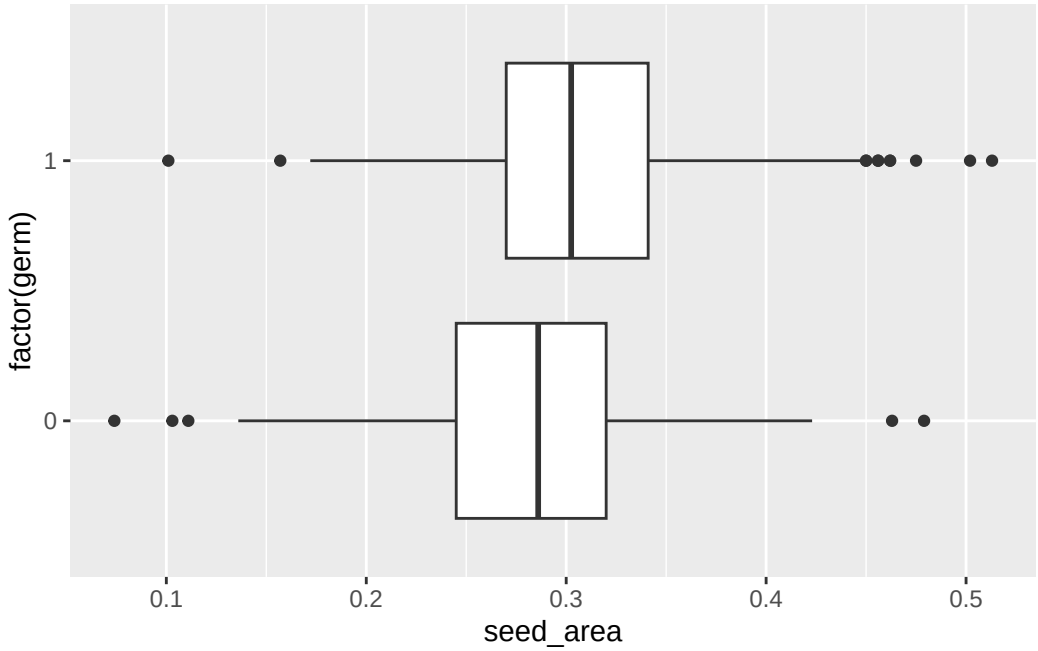

```
pos <- position_jitterdodge(jitter.width = .2, jitter.height = 0)
d %>%
  ggplot(aes(x = mother, y = germ, color = father)) +
  # use bootstrap to plot the means and confidence intervals.
  stat_summary(fun.data = mean_cl_boot, geom = "point",
               position = pos) +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar",
               position = pos)
```

```
Warning: Computation failed in `stat_summary()`.
Computation failed in `stat_summary()`.
Caused by error in `fun.data()`:
! The package "Hmisc" is required.
```



This is ugly, but I don't want to spend more time on it now. The pattern looks very much like the same plot made with seed size, so I'm expecting seed size should be predictive of germination. Let's look at that next.

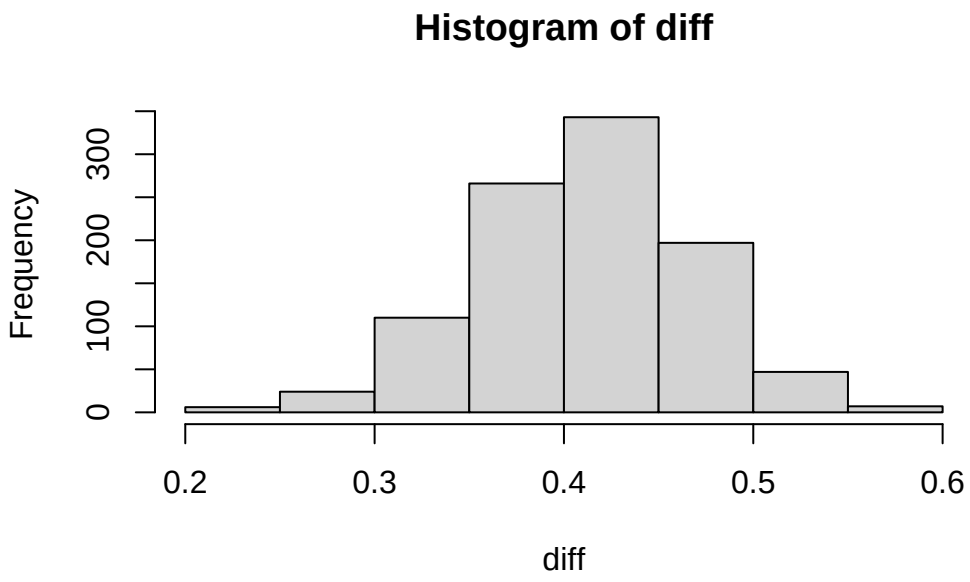
```
d %>%
  ggplot(aes(x = seed_area, y = factor(germ))) +
  geom_boxplot()
```



Sure looks like the germinants were bigger. Let's do just a quick bootstrap to test the difference in the means.

```
germ <- d$germ
# standardize seed area for ease of interpretation
seed_area <- (d$seed_area - mean(d$seed_area))/sd(d$seed_area)
diff <- c()
for(i in 1:1000){
  samps <- sample(1:nrow(d), replace = T)
  g <- germ[samps]
  s <- seed_area[samps]
  diff[i] <- mean(s[g==1]) - mean(s[g==0])
}

hist(diff)
```



So the germinating seeds are roughly 0.4 standard deviations larger than the non-germinants–on average.

Seedling Sizes

The next set of data that needs to be wrangled is the seedling area measurements. At each timepoint, the seedlings were photographed and measured by two people (or will be, I still need to finish my measurements…). It might not be the absolute best approach—probably should model the differences—but I'm going to average across the multiple observations of each seedling at each timepoint.

```
seedlings <- read.csv(here("data/raw/seedling_areas.csv"))

seedlings %>%
  # make area numeric, then drop the NAs to get just observations.
  mutate(area = as.numeric(area)) %>%
  drop_na(area) %>%
  # now average each seedling time.
  group_by(tray, block, ind, time) %>%
  summarise(area = round(mean(area),3)) %>%
  ungroup() %>%
  mutate(tray = paste0("lc_", tray)) %>%
```

```
  # the time stamp is the same as above for germination, just going to
  # make it the day
  separate(time, into = c("day", NA), sep = "_") %>%
  mutate(day = gsub("24", "", day),
         day = as.numeric(day)) %>%
  # join with the germ day data to get cross information as well
  # as germination time to calculate age.
  left_join(germ_day, by = join_by(tray, block, ind)) %>%
  mutate(age = day - germ_day) %>%
  select(tray, block, ind, area, age, cross, mother, father)
```

```
Warning: There was 1 warning in `mutate()`.
i In argument: `area = as.numeric(area)`.
Caused by warning:
! NAs introduced by coercion
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, ind, and time.
i Output is grouped by tray, block, and ind.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind, time))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 5,505 x 8
   tray  block   ind   area   age cross mother father
   <chr> <int> <int>  <dbl> <dbl> <fct> <chr>  <chr>
 1 lc_1      1     1  0.989     1 f2    B12_1  B12_11
 2 lc_1      1     1  4.50      4 f2    B12_1  B12_11
 3 lc_1      1     1  6.81      6 f2    B12_1  B12_11
 4 lc_1      1     1  8.93      8 f2    B12_1  B12_11
 5 lc_1      1     1 14.8      11 f2    B12_1  B12_11
 6 lc_1      1     1 18.3      12 f2    B12_1  B12_11
 7 lc_1      1     2  5.92      3 f2    B12_1  B12_11
 8 lc_1      1     2  8.42      5 f2    B12_1  B12_11
 9 lc_1      1     2 12.6       7 f2    B12_1  B12_11
10 lc_1      1     2 24.0      10 f2    B12_1  B12_11
# i 5,495 more rows
```

Add the above code to `wrangling.R` to create the seedling area file.